

## KUCOVE, Albania

WMO: 136240

Lat: 40.800N

Lon: 19.900E

Elev: 108

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>27.2 | (c)<br>30.1 | (d)<br>15.4                   | (e)<br>12.0 | (f)<br>37.5 | (g)<br>19.8 | (h)<br>15.0 | (i)<br>41.0 | (j)<br>21.5           | (k)<br>54.7 | (l)<br>18.2 | (m)<br>53.3 | (n)<br>3.3            | (o)<br>140 | (p)<br>0.388 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>24.4            | (c)<br>96.8     | (d)<br>72.3 | (e)<br>94.1 | (f)<br>71.7 | (g)<br>91.8 | (h)<br>71.3 | (i)<br>75.6         | (j)<br>90.1 | (k)<br>74.3 | (l)<br>88.8 | (m)<br>73.0 | (n)<br>87.5 | (o)<br>5.9           | (p)<br>320 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 71.3                            | 115.9 | 83.1 | 69.8 | 110.0 | 80.9 | 68.3 | 104.3 | 79.9 | 39.2          | 90.0 | 37.8 | 89.1 | 36.6 | 87.7 | 83.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |         |                    |         | n-Year Return Period Values of Extreme Temperature |         |            |         |            |         |            |         |    |    |
|-------------------|----------|----------|----------------------------|---------|--------------------|---------|----------------------------------------------------|---------|------------|---------|------------|---------|------------|---------|----|----|
|                   |          |          | Mean                       |         | Standard Deviation |         | n=5 years                                          |         | n=10 years |         | n=20 years |         | n=50 years |         |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max     | Min                | Max     | Min                                                | Max     | Min        | Max     | Min        | Max     | Min        | Max     | DB | WB |
| (a) 18.5          | (b) 14.9 | (c) 12.6 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |    |    |
| (a) 18.5          | (b) 14.9 | (c) 12.6 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 62.2   | 45.5  | 49.1  | 53.8  | 59.4  | 65.5  | 73.9  | 79.3  | 78.7  | 72.8  | 64.1  | 56.0  | 47.9  |
|                                                                        | DBStd       | 12.61  | 6.01  | 5.38  | 4.48  | 4.75  | 5.07  | 5.54  | 3.69  | 3.96  | 4.99  | 5.01  | 5.73  | 5.44  |
|                                                                        | HDD50       | 373    | 162   | 71    | 15    | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 16    | 106   |
|                                                                        | HDD65       | 2518   | 605   | 445   | 349   | 178   | 55    | 3     | 0     | 0     | 5     | 75    | 273   | 530   |
|                                                                        | CDD50       | 4840   | 22    | 47    | 132   | 285   | 481   | 717   | 907   | 889   | 685   | 438   | 196   | 41    |
|                                                                        | CDD65       | 1510   | 0     | 1     | 1     | 10    | 72    | 270   | 442   | 424   | 240   | 48    | 2     | 0     |
|                                                                        | CDH74       | 15712  | 0     | 2     | 14    | 124   | 634   | 2772  | 4996  | 4545  | 2182  | 433   | 10    | 0     |
|                                                                        | CDH80       | 7359   | 0     | 0     | 1     | 21    | 138   | 1214  | 2639  | 2390  | 898   | 58    | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 4.8    | 5.4   | 6.4   | 6.2   | 4.2   | 4.1   | 4.2   | 4.1   | 3.9   | 4.9   | 3.9   | 5.5   | 5.1   |
| (15) Precipitation                                                     | PrecAvg     | 39.9   | 3.6   | 4.1   | 3.6   | 3.5   | 2.2   | 1.2   | 0.9   | 1.1   | 3.3   | 4.4   | 5.7   | 5.5   |
|                                                                        | PrecMax     | 58.3   | 7.4   | 9.7   | 7.5   | 6.1   | 5.7   | 2.9   | 2.8   | 3.7   | 11.1  | 9.4   | 9.5   | 14.1  |
|                                                                        | PrecMin     | 28.5   | 0.6   | 0.1   | 0.3   | 1.6   | 0.7   | 0.3   | 0.0   | 0.0   | 0.2   | 0.6   | 1.3   | 0.2   |
|                                                                        | PrecStd     | 8.3    | 2.0   | 2.5   | 2.1   | 1.2   | 1.2   | 0.7   | 0.7   | 1.0   | 2.6   | 2.4   | 2.3   | 3.5   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 64.4  | 71.3  | 75.6  | 83.6  | 87.3  | 96.3  | 100.2 | 100.4 | 95.2  | 84.4  | 75.3  | 66.1  |
|                                                                        |             | MCWB   | 52.4  | 54.2  | 55.3  | 63.6  | 67.7  | 72.3  | 74.4  | 72.4  | 70.7  | 63.6  | 61.0  | 55.2  |
|                                                                        | 2%          | DB     | 61.2  | 66.6  | 71.3  | 77.9  | 84.0  | 93.4  | 97.3  | 96.6  | 91.6  | 81.2  | 71.5  | 62.9  |
|                                                                        |             | MCWB   | 51.2  | 53.0  | 54.8  | 61.2  | 66.4  | 71.9  | 73.4  | 71.5  | 69.0  | 64.2  | 58.9  | 53.3  |
|                                                                        | 5%          | DB     | 58.5  | 62.7  | 68.2  | 74.3  | 80.6  | 90.2  | 94.2  | 93.8  | 88.5  | 78.9  | 69.0  | 60.3  |
|                                                                        |             | MCWB   | 49.7  | 51.0  | 54.5  | 59.4  | 65.1  | 70.9  | 72.5  | 71.1  | 67.8  | 63.9  | 57.8  | 51.8  |
|                                                                        | 10%         | DB     | 55.5  | 59.8  | 64.9  | 71.2  | 77.3  | 87.0  | 91.6  | 91.2  | 85.4  | 75.9  | 66.5  | 57.7  |
|                                                                        |             | MCWB   | 48.3  | 49.6  | 53.1  | 58.3  | 63.9  | 69.8  | 71.7  | 71.3  | 67.2  | 62.7  | 56.7  | 50.3  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 54.5  | 56.5  | 59.8  | 65.9  | 70.8  | 76.1  | 79.2  | 76.8  | 73.6  | 69.3  | 63.6  | 57.3  |
|                                                                        |             | MCDB   | 61.2  | 67.6  | 70.1  | 79.5  | 81.4  | 89.9  | 94.3  | 90.7  | 86.9  | 78.7  | 70.3  | 63.2  |
|                                                                        | 2%          | WB     | 52.4  | 54.1  | 57.1  | 63.0  | 68.6  | 74.1  | 76.3  | 75.2  | 71.2  | 67.2  | 61.1  | 55.0  |
|                                                                        |             | MCDB   | 59.1  | 62.6  | 67.3  | 74.4  | 80.0  | 88.2  | 91.1  | 89.5  | 85.6  | 77.0  | 68.9  | 60.4  |
|                                                                        | 5%          | WB     | 50.9  | 52.6  | 55.6  | 61.3  | 66.6  | 72.5  | 74.5  | 73.9  | 70.0  | 65.4  | 59.4  | 53.3  |
|                                                                        |             | MCDB   | 57.1  | 60.7  | 65.4  | 71.6  | 78.0  | 86.0  | 89.3  | 88.4  | 83.0  | 75.0  | 66.9  | 58.4  |
|                                                                        | 10%         | WB     | 49.2  | 50.9  | 54.1  | 59.5  | 64.9  | 70.8  | 73.1  | 72.7  | 68.7  | 63.6  | 57.6  | 51.4  |
|                                                                        |             | MCDB   | 54.6  | 58.5  | 63.2  | 68.9  | 75.0  | 84.3  | 88.2  | 87.1  | 81.6  | 72.9  | 64.4  | 56.3  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 19.5  | 19.2  | 20.5  | 21.0  | 20.6  | 22.5  | 24.4  | 25.3  | 24.5  | 22.9  | 19.3  | 18.9  |
|                                                                        |             | MCDBR  | 21.4  | 23.5  | 26.8  | 26.4  | 25.0  | 26.1  | 27.8  | 27.6  | 27.9  | 26.6  | 22.5  | 21.2  |
|                                                                        |             | MCWBR  | 12.6  | 12.3  | 12.9  | 12.3  | 10.6  | 9.4   | 9.5   | 8.8   | 9.6   | 12.3  | 11.6  | 12.8  |
|                                                                        | 5% WB       | MCDBR  | 18.4  | 20.8  | 22.3  | 22.6  | 22.5  | 23.6  | 25.2  | 24.8  | 24.1  | 23.0  | 18.9  | 17.3  |
|                                                                        |             | MCWBR  | 11.6  | 11.8  | 12.2  | 12.3  | 11.0  | 10.0  | 11.3  | 10.5  | 10.5  | 12.8  | 11.1  | 11.6  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.340 | 0.367 | 0.413 | 0.466 | 0.456 | 0.433 | 0.438 | 0.454 | 0.434 | 0.401 | 0.367 | 0.342 |
|                                                                        | taud        |        | 2.426 | 2.328 | 2.194 | 2.060 | 2.152 | 2.266 | 2.250 | 2.199 | 2.246 | 2.319 | 2.375 | 2.423 |
|                                                                        | Ebn at Noon |        | 257   | 265   | 263   | 257   | 262   | 268   | 266   | 257   | 254   | 249   | 242   | 245   |
|                                                                        | Edh at Noon |        | 26    | 33    | 42    | 51    | 48    | 43    | 43    | 44    | 39    | 33    | 27    | 24    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 582   | 826   | 1218  | 1574  | 1954  | 2256  | 2327  | 2045  | 1499  | 1046  | 666   | 517   |
|                                                                        | RadStd      |        | 62    | 115   | 153   | 157   | 132   | 130   | 98    | 105   | 105   | 101   | 61    | 88    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (18 neighbors) | +0.65 | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | -149  | N/A   | N/A   |

Nomenclature: See separate page